

9.4.4.

In a group of 700 people, must there be 2 who have the same first and last initials? Why?

There are 26 letters of the alphabet, and we are looking for the same first and last initial for 700 people. Then there are 26 possibilities for the first initial, and 26 possibilities for the last initial.

$$(A, A), (A, B), (A, C), \dots, (Z, Z)$$

$$26 \times 26 = 26^2 = 676$$

Let $f : \{700 \text{ people}\} \rightarrow \{676 \text{ initial pairs}\}$, then the function cannot be injective, because the domain has more elements than the codomain. Hence $\exists x_1, x_2, x_1 \neq x_2$ such that $f(x_1) = f(x_2)$. Therefore, the pigeonhole principle guarantees that at least two people have the same first and the same last initial.

Solution: Since 700 people $>$ 676 initial pairs, then the pigeonhole principle guarantees that at least two people have the same first and the same last initial.